NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 23/01/2026
Time: 3 Hrs | GTM – 23 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	B	A	C	B	D	A	A	C	D
11	12	13	14	15	16	17	18	19	20
B	B	A	A	A	B	C	A	A	D
21	22	23	24	25					
0	1	1	0	10					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	B	B	A	D	C	B	C	A	A
36	37	38	39	40	41	42	43	44	45
B	C	B	A	C	A	D	C	C	D
46	47	48	49	50					
25	104	54	36	56					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	C	B	C	D	C	B	A	C	B
61	62	63	64	65	66	67	68	69	70
D	B	C	C	D	C	C	D	A	A
71	72	73	74	75					
18	35	7	30	37					

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HINTS
PHYSICS

1. Since electrostatic energy stored is

$U = \int \frac{1}{2}\varepsilon_0 E^2 \mathrm{dv}$ where dr is small volume

$\Rightarrow U = \frac{1}{2}\varepsilon_0 E^2 \int d\mathrm{v}$, where $E = \frac{\sigma}{2\varepsilon_0} = \text{constant}$

$$\Rightarrow U = \frac{1}{2}\varepsilon_0 \left(\frac{\sigma^2}{4\varepsilon_0^2}\right) a^3 = \frac{\sigma^2 a^3}{8\varepsilon_0}$$

2. Here $P_1 = 6\ \mathrm{D}$, $P_2 = -2\ \mathrm{D}$

Power of the combination is given by

$$P = P_1 + P_2 = 6 - 2 = 4\ \mathrm{D}$$

Since $f = \frac{1}{P}$

$$\Rightarrow \quad f = \frac{1}{4} = 0.25\ \mathrm{m} = 25\ \mathrm{cm}$$

3. Surface tension of water $T = 7.2 \times 10^{-2}\ \mathrm{Nm}^{-1}$

Radius of air bubble $R = 0.1\ \mathrm{mm} = 10^{-4}\ \mathrm{m}$

The excess pressure inside the air bubble is given by,

$$P_2 - P_1 = \frac{2T}{R}$$

Pressure inside the air bubble is

$$P_2 = P_1 + \frac{2T}{R}$$

Substituting the values, we have

$$P_2 = (1.013 \times 10^5) + \frac{(2 \times 7.2 \times 10^{-2})}{10^{-4}}$$

$$\Rightarrow \quad P_2 = 1.027 \times 10^5\ \mathrm{Nm}^{-2}$$

4. Volume of cylinder is given by

$$V = \frac{\pi D^2 h}{4}$$

$$\Rightarrow \quad V = \frac{\pi}{4} \times 34.2 \times (12.6)^2 = 4264.39\ \mathrm{cm}^3$$

Since, $\frac{\Delta V}{V} = \frac{2\Delta D}{D} + \frac{\Delta h}{h}$

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$$\Rightarrow \quad \frac{\Delta V}{V} = \frac{2\times 0.1}{12.6} + \frac{0.1}{34.2}$$

$$\Rightarrow \quad \Delta V = 80.157$$

Reducing the answers to appropriate significant figures, then

$$V = 4260 \pm 80 \text{ cm}^3$$

Hence, the correct answer is C.

5.

Since, we know that, in ground state, $n = 1$, we have

$$TE = -KE = \frac{PE}{2}$$

$$\Rightarrow \quad PE = 2(TE) = 2(-13.6 \text{ eV})$$

$$\Rightarrow \quad PE = -27.2 \text{ eV}$$

However, we have assumed this energy to be zero i.e., potential energy is increased by 27.2 eV . Since, kinetic energy in all energy states will remain unchanged whereas potential energy and hence, the total energy in all states will increase by 27.2 eV . Further, first excited state means $n = 2$, so

$$E_2 = -3.4 \text{ eV (previously)}$$

$$\Rightarrow \quad E_2' = -3.4 + 27.2 = 23.8 \text{ eV (now)}$$

6. Conceptual

7.

Let r be the radius of smaller drops and R of bigger one. Equating the initial and final volumes, we have,

$$\frac{4}{3}\pi R^3 = (1000)\left(\frac{4}{3}\pi r^3\right)$$

$$\Rightarrow \quad R = 10r = (10)(10^{-7}) \text{ m}$$

$$\Rightarrow \quad R = 10^{-6} \text{ m}$$

Further, the water drops have only one free surface. Therefore,

$$\Delta A = 4\pi R^2 - (1000)(4\pi r^2)$$

$$\Rightarrow \quad \Delta A = 4\pi\left[(10^{-6})^2 - (10^3)(10^{-7})^2\right]$$

$$\Rightarrow \quad \Delta A = -36\pi(10^{-12}) \text{ m}^2$$

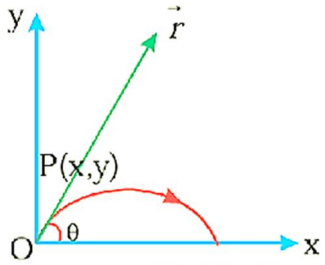
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Here, negative sign implies that surface area is decreasing.
Hence, energy released in the process.

$$\Delta U = T|\Delta A| = (7\times10^{-2})(36\pi\times10^{-12})\ \text{J}$$

$$\Rightarrow \quad \Delta U = 7.9\times10^{-12}\ \text{J}$$

8. Velocity at any time 't' is $\vec{v} = v_x\hat{i} + v_y\hat{j}$



Position vector of projectile at time 't' is $\vec{r} = x\hat{i} + y\hat{j}$

Since $\vec{L} = m(\hat{r}\times\hat{v})$

$$\vec{L} = m\left[\left(x\hat{i}+y\hat{j}\right)\times\left(v_x\hat{i}+v_y\hat{j}\right)\right]$$

$$\vec{L} = m\left[xv_y\hat{k} - yv_x\hat{k}\right]; \vec{L} = m\vec{k}\left[xv_y - yv_x\right]$$

$$v_x = \frac{u}{\sqrt{2}} \text{ and } x = \frac{u^2}{\sqrt{2}g}$$

Where $v_x = \frac{u}{\sqrt{2}}$ and $x = \frac{u^2}{\sqrt{2}g}$

$$\left(\because x = \left(u\cos45^0\right)t = \frac{u}{\sqrt{2}}\left(\frac{u}{g}\right)\right)$$

$$v_y = u\sin45^0 - gt = \frac{1-\sqrt{2}}{\sqrt{2}}v_0$$

$$y = u\sin45^0 t - \frac{1}{2}gt^2 = \frac{y^2}{2g}\left(\sqrt{2}-1\right); \vec{L} = \frac{-\hat{k}mu^3}{2\sqrt{2}g}$$

9. Let n_1 maxima of λ_1 coincides with n_2 maxima of λ_2. Then, $y_{n_1} = y_{n_2}$

$$\Rightarrow \quad \frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d}$$

$$\Rightarrow \quad \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{5200}{6500} = \frac{4}{5}$$

$$\Rightarrow \quad 4\lambda_1 = 5\lambda_2$$

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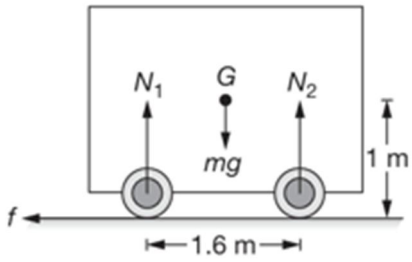
Thus, fourth maxima of λ_1 coincides with fifth maxima of λ_2. The minimum value of $y(\neq 0)$ is given by

$$y = \frac{4\lambda_1 D}{d} = \frac{4(0.65\times10^{-6})(2)}{6.5\times10^{-3}} = 0.8 \text{ mm}$$

10.

The situation is shown in figure. Let N_1 and N_2 be the reactions at inner wheels and outer wheels respectively. f is the frictional force of the tracks. G represents the centre of gravity and mg, the weight of the vehicle acting downwards at centre of gravity. For vertical equilibrium,

$$N_1 + N_2 = mg \qquad \ldots(1)$$



For circular motion

$$f = \frac{mv^2}{r}$$

For rotational equilibrium, net moment of all the forces about G should be zero. Hence,

$$f(1) + N_1\left(\frac{1.6}{2}\right) = N_2\left(\frac{1.6}{2}\right)$$

$$\Rightarrow \quad 0.8N_1 + f = 0.8N_2 \qquad \ldots(3)$$

When the inner wheel leaves the road, then $N_1 = 0$. Therefore from (3), we get

$$f = 0.8N_2 \qquad \ldots(4)$$

and from equation (1) we get

$$N_2 = mg \qquad \ldots(5)$$

Solving equations (2), (4) and (5), we get

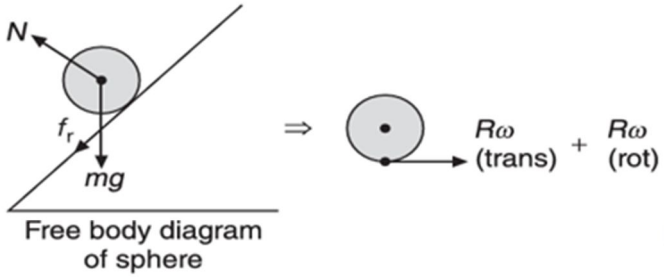
$$v = \sqrt{(0.8)\times(50)\times(10)} = 20 \text{ ms}^{-1}$$

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11.

Velocity of lower most point is not zero, hence it is not the case of pure rolling and friction between the sphere and the incline is kinetic in nature. Since velocity of lower most point in forward direction therefore friction will act in backward direction.

Let retardation is a and angular retardation is α



$$\Rightarrow \quad mg\sin\theta + \mu mg\cos\theta = ma$$

$$\Rightarrow \quad a = g(\sin\theta + \mu\cos\theta) = \frac{g}{\sqrt{2}}(1+0.5) = \frac{1.5g}{\sqrt{2}}$$

Now, τ due to friction is

$$\tau = R(\mu mg\cos\theta)$$

$$\Rightarrow \quad \alpha = \frac{(r\mu mg\cos\theta)}{\frac{2}{5}mR^2} = \frac{5g\mu\cos\theta}{2R}$$

$$\Rightarrow \quad \alpha = \frac{5g(0.5)(1/\sqrt{2})}{2R}$$

$$\Rightarrow \quad \alpha = \frac{5g}{4R\sqrt{2}}$$

For translatory motion

$$v' = v - at = R\omega - at$$

When $v' = 0$, then

$$t = \frac{R\omega}{a} = \frac{R\omega\sqrt{2}}{1.5g} = \frac{\sqrt{2}}{1.5}\left(\frac{R\omega}{g}\right)$$

For rotatory motion, we have

$$\omega' = \omega - \alpha t$$

$$\Rightarrow \quad t = \frac{\omega}{\alpha} = \frac{\omega}{\frac{5g}{4R\sqrt{2}}} = \frac{4\sqrt{2}R\omega}{5g}$$

12. As according to lens – makers' formula

$$\frac{f_w}{f_a} = \frac{(\mu_g - 1)}{\left(\frac{\mu_g}{\mu_w} - 1\right)}; \frac{f_w}{0.3} = \frac{\left(\frac{3}{2} - 1\right)}{\left(\frac{3/2}{4/3} - 1\right)} \Rightarrow f_w = 1.2m$$

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13.

Let $x_1 = a\sin\omega t$ and $x_2 = a\sin(\omega t+\phi)$ be two SHMs.

$$\Rightarrow \quad \frac{a}{3} = a\sin\omega t \text{ and } -\frac{a}{3} = a\sin(\omega t+\phi)$$

$$\Rightarrow \quad \sin\omega t = \frac{1}{3} \text{ and } \sin(\omega t+\phi) = -\frac{1}{3}$$

$$\Rightarrow \quad \frac{1}{3}\cos\phi + \sqrt{1-\frac{1}{9}}\sin\phi = -\frac{1}{3}$$

$$\Rightarrow \quad 9\cos^2\phi + 2\cos\phi - 7 = 0$$

$$\Rightarrow \quad \cos\phi = \frac{-2\pm\sqrt{4+4\times7\times9}}{18} = -1,\ \frac{7}{9}$$

$$\Rightarrow \quad \delta = 180^\circ \text{ or } \delta = \cos^{-1}\left(\frac{7}{9}\right)$$

Now $v_1 = a\omega\cos\omega t$, $v_2 = a\omega\cos(\omega+\phi)$. When we substitute $\phi = 180^\circ$ we find that v_1 and v_2 are of opposite signs. Hence $\phi = 180^\circ$ is not acceptable.

$$\Rightarrow \quad \phi = \cos^{-1}\left(\frac{7}{9}\right)$$

14.

$$E_i = -\frac{GMm}{2r},\ E_f = -\frac{GMm}{2R},\ E_i > E_f$$

$$\Rightarrow \quad \Delta E = E_i - E_f = \frac{GMm}{2}\left(\frac{r-R}{Rr}\right)$$

$$\Rightarrow \quad t = \frac{\Delta E}{W}$$

$$\Rightarrow \quad t = \frac{GMm}{2W}\left(\frac{1}{R}-\frac{1}{r}\right)$$

15.

Let f_1 and f_2 be the focal lengths in water.

Then

$$\frac{1}{f_1} = \left(\frac{\mu_1}{\mu_w}-1\right)\left(\frac{1}{R}+\frac{1}{R}\right) \Rightarrow \frac{1}{f_1} = \left(\frac{\mu_1}{\mu_w}-1\right)\left(\frac{2}{R}\right) \rightarrow (1)$$

$$\frac{1}{f_2} = \left(\frac{\mu_2}{\mu_w}-1\right)\left(-\frac{1}{R}-\frac{1}{R}\right)$$

$$\frac{1}{f_2} = \left(\frac{\mu_2}{\mu_w}-1\right)\left(-\frac{2}{R}\right) \rightarrow (2)$$

Adding Eqs. (1) and (2) we get

$$\frac{1}{f_1}+\frac{1}{f_2} = \frac{2(\mu_1-\mu_2)}{\mu_w R}$$

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But the given system is equal to combination of two lens kept in contact in liquid so

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2} \text{ or } \frac{1}{30}=\frac{2(\mu_1-\mu_2)}{\mu_w R}$$

$\therefore (\mu_1-\mu_2)=\frac{\mu_w R}{60}$; substituting the values we get $(\mu_1-\mu_2)=\frac{4\times 1}{3\times 60}=\frac{1}{3}$

16.

$i = i_1+i_2 = 3+4\sqrt{2}\sin\omega t$

$$i_{rms}^2=\frac{\int_0^T i^2\,dt}{\int_0^T dt}=\frac{\int_0^T\left(3+4\sqrt{2}\sin\omega t\right)^2 dt}{T}$$

$$i_{rms}^2=\frac{1}{T}\int_0^T\left(9+24\sqrt{2}\sin\omega t+32\sin^2\omega t\right)dt$$

$\therefore i_{rms} = 5A$

17.

Kinetic energy gained by a charge q after being accelerated through a potential difference V volt is

$$qV=\frac{1}{2}mv^2$$

$$\Rightarrow \quad v=\sqrt{\frac{2qV}{m}}$$

$$\Rightarrow \quad mv=\sqrt{2mqV}$$

So, the de Broglie wavelength is given by

$$\lambda=\frac{h}{mv}=\frac{h}{\sqrt{2mqV}}$$

$$\Rightarrow \quad \frac{\lambda_p}{\lambda_\alpha}=\sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}}$$

$$\Rightarrow \quad \frac{\lambda_p}{\lambda_\alpha}=\sqrt{\frac{4\times 2}{1\times 1}}=2\sqrt{2}$$

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18.

Potential difference between the conductors is

given by $V = V_+ - V_- = \int_0^X E dx$

$$\Rightarrow V = \int_0^X \frac{Q}{\epsilon_0 A}(1 + Bx^2)dx$$

$$\text{or } V = \frac{Q}{\epsilon_0 A}\left(x + \frac{Bx^3}{3}\right)_0^X = \frac{Q}{\epsilon_0 A}\left(X + \frac{BX^3}{3}\right)$$

$$\text{Capacity } C = \frac{Q}{V} = \frac{\epsilon_0 A}{X\left(1 + \frac{BX^2}{3}\right)}$$

19.

$$W = \int_A^B \vec{F} \cdot d\vec{r}\text{, where } d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$$

$$\Rightarrow \quad W = \int_A^B \left(y\hat{i} + x\hat{j}\right) \cdot \left(\hat{i}dx + \hat{j}dy + \hat{k}dz\right)$$

$$\Rightarrow \quad W = \int_A^B \left(ydx + xdy\right) = \int_A^B d\left(xy\right)$$

$$\Rightarrow \quad W = \left(xy\right)\Big|_{(0,0,0)}^{(1,1,1)}$$

$$\Rightarrow \quad W = 1\,\text{J}$$

20. Conceptual

21.

The electric field due to Q at any point of the square will be along the plane of square and the electric field line are perpendicular to square, so $\phi = 0$.
In other words we can say that no line is crossing the square so flux $= 0$.

22. Conceptual
23. Conceptual
24. Conceptual

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25.

Given: 2 moles of a monoatomic ideal gas, isobaric heating by $\Delta T = 40^{\circ}$C. Heat added $= Q_1$, change in internal energy $= \Delta U = Q_2$. For an ideal gas:

$$Q_1 = nC_p\Delta T, \qquad Q_2 = nC_v\Delta T$$

So

$$\frac{Q_1}{Q_2} = \frac{C_P}{C_V} = \gamma$$

For a monoatomic ideal gas:

$C_P = C_V + r = \frac{5}{2}R$ Thus $C_v = \frac{3}{2}R$'

$$\gamma = \frac{C_P}{C_V} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3}.$$

We are told $\frac{Q_1}{Q_2} = \frac{x}{6}$. Therefore

$$\frac{5}{3} = \frac{x}{6} \Rightarrow x = 6.\frac{5}{3} = 10$$

CHEMISTRY

26. $Zn + \underset{(0.1)}{H^+} \rightarrow \underset{(0.01)}{Zn^{2+}} + H_2 \uparrow$

$$E_{cel} = 0.77 + \log_{10}\left[\frac{0.01}{(0.1)^2}\right]$$

$$E_{cel} = 0.77V$$

27. Number of spectral lines $= \frac{n(n+1)}{2}$

$n = n_2 - n_1$

28. Millimoles of $AgNO_3 = \frac{1.2\times10^{22}}{6\times10^{23}}\times1000 = 20$

Millimoles of $CoCl_3.6H_2O = 0.1\times100 = 10$

Each $CoCl_3.6H_2O$ gives two chloride ions.

$\left[Co(H_2O)_5\,Cl\right]Cl_2H_2O$

29. More resonance stabilised.

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30. Conceptual

31. Cu^{2+} has a hydration enthalpy of $2121\ J/mole$ which accounts for its positive $E^0_{M^{2+}/M}$ value

32. $K_{sp} = 4S^3$

33.
$\Delta G = \Delta H - T\Delta S$
$For\ equilibrium\ \Delta G = 0$
$For\ spon\tan eous\ reaction, \Delta G < 0$
$\therefore T > T_e$

34. $n = \dfrac{M.wt}{Emp.wt}$
$M.f = n(Emp\ for)$

35. In ClO_4^- ion oxidation state of $'Cl'$ is maximum i.e., +7, so $'Cl'$ cannot show more than +7 oxidation state

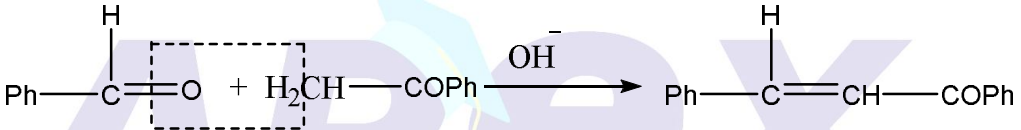
36. $sugar\ A \rightarrow RNA$
$sugar\ B \rightarrow DNA$

37. $KMnO_4$ cannot reduce H_2SO_4 to SO_2

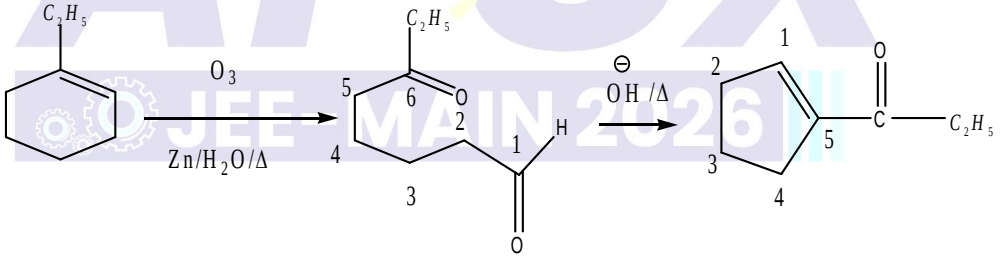
38. $l_1 = 204\,pm,\ \theta_1 = 107^0$
$l_2 = 205.7\,pm,\ \theta_2 = 102.2^0$

39. $K_p = \dfrac{4\alpha^2 P}{1-\alpha^2}$

40.



41.



42. Conceptual

43. $\psi_A - \psi_B = \psi_{MO}$ is anti-bonding molecular orbital
$\mu = Q \times r$ IS Dipole moment

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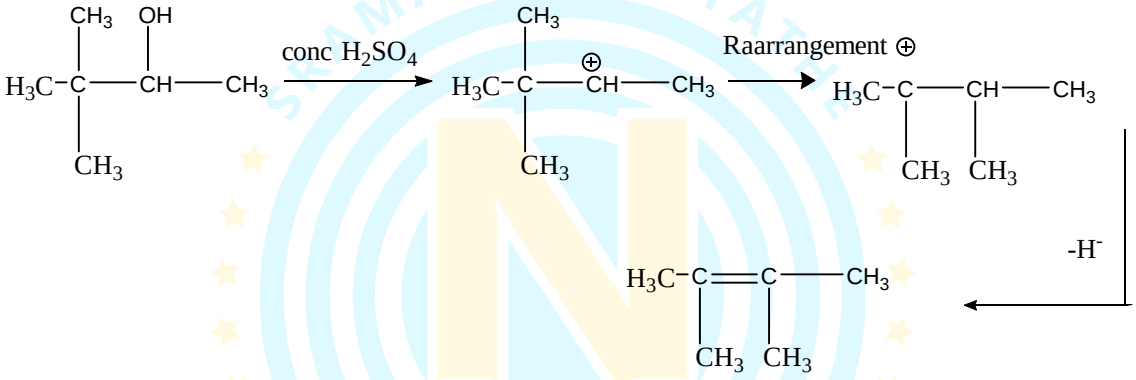
$\frac{N_b - N_a}{2} = bonding\ order$

$\psi_{MO} = \psi_A + \psi_B$ is bonding molecular orbital

44. $A = C_2H_2 \quad B = C_6H_6 \quad C = C_6H_{12}\left(\text{cyclohexane}\right)$

$D = C_6H_6Cl_6\left(\text{hexa chloro cyclohexane}\right)$

45.



46. $K_P = P_{CO_2} = 2.16$

$PV = nRT$

$\frac{2.16 \times 10}{0.082 \times 1073} = 0.2454$

$\%dissociation = \frac{0.2454}{1} \times 100 = 24.54\% \simeq 24\ \left(round\ off\right)$

47. Enthalpy of combustion of propane, graphite and hydrogen at 298K are

$C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(l).} \quad \Delta H_1 = -2220 KJmol^{-1}$

$C_{(graphite)} + O_{2(g)} \rightarrow CO_{2(g)}.\Delta H_2 = -393.5 KJmol^{-1}$

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}.\Delta H_3 = -285.8 KJmol^{-1}$

The reaction is $3C_{(graphite)} + 4H_{2(g)} \rightarrow C_3H_{8(g)}$

$\Delta H_1 = 3\Delta H_2 + 4\Delta H_3 - \Delta H_1$

$= 3(-393.5) + 4(-285.8) - (-2220)$

$= -103.7 KJmol^{-1}$

$\left|\Delta H_1\right| = 104 KJmol^{-1}$

48. No of moles initial present $MV = (0.5)(0.8) = 0.4\,mol$

No of moles evaporated $= \frac{11.5}{63} = 0.182\ mol$

No of moles left in the final solution $= 0.4 - 0.182 = 0.218\,mol$

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Final volume of solution $= \frac{800}{2} = 400ml = 0.4lit$

Final molarity $= \frac{n}{v} = \frac{0.218}{0.4} = 0.545 = 54.5 \times 10^{-2}$

$= 54$

49. $X = 6,\ X^2 = 36$

$CdS \rightarrow Yellow,\ As_2S_3 \rightarrow Yellow,$

$As_2S_5 \rightarrow Yellow, PbI_2 - Yellow,$

$SnS_2 \rightarrow Yellow,\ AgI \rightarrow Yellow$

50.

$M_2O_3 + 3H_2 \rightarrow 2M + 3H_2O$

$2(M)+3(16) \quad 6$

$? \quad 1.2$

$X = \frac{12(2(M)+98)}{6} = 0.2((2M)+48) = 32$

$2(M)+48 = 160$

$2(M) = 112$

$M = 56gm\ (\because M = Atomic\ weight\ of\ element)$

MATHEMATICS

51. Solving y=7x+5 and the circle $x^2+y^2=1$ $A\left(-\frac{3}{5},\frac{4}{5}\right)$ and $B\left(-\frac{4}{5},-\frac{3}{5}\right)$

$m_{OA} = -\frac{4}{3}$; $m_{OB} = \frac{3}{4}$ Hence, $\angle AOD = 90^0 \Rightarrow \angle ACB = \frac{\pi}{4}$

52. $(x_1 \pm lr,\ y_1 \pm mr,\ z_1 \pm nr)\ ; (2-\frac{1}{3}(6), -3-\left(\frac{-2}{3}\right)6, -5-6\left(\frac{-2}{3}\right)\ ; \Rightarrow (0,1,-1)$

53. Let the equation of secant be $\frac{x-2}{\cos\theta} = \frac{y-2}{\sin\theta} = r$ $(parametric\ form)$ solving it with ellipse, we get $\frac{(r\cos\theta+2)^2}{25} + \frac{(r\sin\theta+2)^2}{16} = 1$;

$r^2(16\cos^2\theta + 25\sin^2\theta) + r(64\cos\theta + 100\sin\theta) - 236 = 0$;

$|r_1 r_2| = PA.PB = \left|\frac{-236}{(16\cos^2\theta + 25\sin^2\theta)}\right| = \left|-\frac{236}{16+9\sin^2\theta}\right|$; Maximum value of PA.PB$= \frac{236}{16} = \frac{59}{4}$;

54. $3l + m + 5n = 0$

$6mn - 2nl + 5ml = 0$

Substituting the value of n from (i) to (ii), then

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$6l^2+9lm-6m^2=0$

$\therefore \frac{l_1}{m_1}=\frac{1}{2}\ and\ \frac{l_2}{m_2}=-2$

From (i), we get

$\frac{l_1}{n_1}=-1\ and\ \frac{l_2}{n_2}=-2$

$\therefore \frac{l_1}{1}=\frac{m_1}{2}=\frac{n_1}{-1}=\sqrt{\frac{l_1^2+m_1^2+n_1^2}{1+4+1}}=\frac{1}{\sqrt{6}}$

And $\frac{l_2}{2}=\frac{m_2}{-1}=\frac{n_2}{-1}=\frac{\sqrt{l_2^2+m_2^2+n_2^2}}{\sqrt{4+1+1}}=\frac{1}{\sqrt{6}}$

If θ be the angle between the lines, then

$\therefore \cos\theta=\left(\frac{1}{\sqrt{6}}\right)\left(\frac{2}{\sqrt{6}}\right)+\left(\frac{2}{\sqrt{6}}\right)\left(-\frac{1}{\sqrt{6}}\right)+\left(-\frac{1}{\sqrt{6}}\right)\left(-\frac{1}{\sqrt{6}}\right)=\frac{1}{6}$

$\therefore \theta=\cos^{-1}\left(\frac{1}{6}\right)$

55. Orthocentre of triangle BCH is the vertex $A(-1,0)$

56. Roots $\rightarrow \alpha=2w^2,\beta=3+4w,\gamma=3+4w^2,$

$\delta=5-w-w^2$

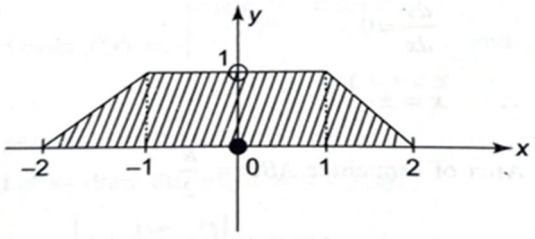
$\delta=5-\left(\omega+\omega^2\right)=5-(-1)=6$

If $\alpha=2w^2$ is a root then $2w$ has to be a root too.

Minimum 5 roots, hence minimum degree $\rightarrow 5$

57. $16\times16-16+(3+4+5)=252\ then\ \bar{x}=\frac{252}{18}=14$

58. $f(x)=\begin{cases}\{x\}, & -2\le x<-1\\ |\operatorname{sgn} x|, & -\le x\le 1\\ \{-x\}, & 1<x\le 2\end{cases}$



The graph of $f(x)$ is

Area $=\frac{2+4}{2}\times1=3\,sq.units$

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59. $f(x)=\lim_{n\to\infty}\dfrac{2x^{2n}\sin\dfrac{1}{x}+x}{1+x^{2n}}$

$$=\begin{cases}2\sin\dfrac{1}{x} & for\ x>1 or\ x<-1\\ \dfrac{2\sin(1/x)+x}{2} & for\ x=1 or -1\\ x & if\ -1<x<1\end{cases}$$

$\lim_{x\to\infty} x\ f(x)=\lim_{x\to\infty}2x\sin\dfrac{1}{x}=2$

$\lim_{x\to1^+} f(x)=2\sin 1;\lim_{x\to1^-} f(x)=1$

$\Rightarrow \lim_{x\to1} f(x)$ *does not exist*

$\lim_{x\to0} f(x)=\lim_{x\to0} x=0$

$\lim_{x\to-\infty} f(x)=\lim_{x\to-\infty}2\sin\dfrac{1}{x}=0$

60. $I_n=\int_0^1 x^{n-1}\,x\sqrt{1-x^2}\ ;\quad \Rightarrow -x^{n-1}\dfrac{(1-x^2)^{3/2}}{3}\Big]_0^1+\int_0^1(n-1)x^{n-2}\ \dfrac{(1-x^2)^{3/2}}{3}\,dx\ ;$

$\dfrac{n-1}{3}I_{n-2}-\dfrac{n-1}{3}I_n\ ;$

$\dfrac{I_n}{I_{n-2}}=\dfrac{n-1}{n+2}\ ;\ \lim_{n\to\infty}\dfrac{I_n}{I_{n-2}}=1\ .$

61. $P(T)=p;P(H)=1-p\quad ;\quad P(T\ or\ HHT\ or\ HHHHT\ or\ .....)\ ;$

$\Rightarrow p+(1-p)^2\ p+(1-p)^4\ p+.....;\Rightarrow\dfrac{p}{1-(1-p)^2}=\dfrac{2}{3}\ ;\Rightarrow p=\dfrac{1}{2}$

62. Put x=2; f (2) + 2 f (1001) =6; put x=1001; f (1001) +2f (2) =3003; f (2) =2000.

63. $(1-(xy)+(xy)^2)dx=x^2dy\ ;\quad let\ xy=v\Rightarrow y=\dfrac{v}{x};\ x^2dy=xdv-vdx\ ;$

$(1-v+v^2)dx=x\,dv-v\,dx\ ;\ \int\dfrac{dv}{1+v^2}=\int\dfrac{dx}{x}\quad \tan^{-1}v=\log x+\log c\ ;$

$xy=\tan\log(cx)$

64. $0\le\sin^2 x\le1;\Rightarrow\left[-\sin^2 x\right]=0\ or\ -1\ ;\quad$ But $\sec^{-1}(0)$ is not defined ;

$y=\sec^{-1}\left[-\sin^2 x\right]=\sec^{-1}(-1)=\pi\ ;\quad now\ y=\dfrac{16-x^2}{4}\Rightarrow x^2=-4(y-4)\ ;$

$required\ area=\int_{-2\sqrt{4-\pi}}^{2\sqrt{4-\pi}}\left(\dfrac{16-x^2}{4}-\pi\right)dx;\ =\dfrac{8}{3}(4-\pi)^{3/2}$

65. $|\vec{a}|=12;|\vec{b}|=4;\vec{a}.\vec{b}=2;\ \vec{c}=(2\vec{a}\times\vec{b})-3\vec{b};$

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$\cos\theta = \frac{2}{|\vec{a}|.|\vec{b}|} = \frac{2}{4} = \frac{1}{2}; \therefore \theta = \frac{\pi}{3}; |\vec{c}+3\vec{b}|^2 = |2\vec{a}\times\vec{b}|^2;$

$|\vec{c}|^2 + 144 + 6\vec{b}.\vec{c} = 48; \vec{b}.\vec{c} = 0 - 3\times 16; \vec{b}.\vec{c} = -48;$

$|\vec{c}|^2 + 96 - 6\times 48 = 0; |\vec{c}|^2 = 192;$

$192 + 96 + 6|\vec{b}|.|\vec{c}| \cos\alpha = 0; \cos\alpha = -\frac{\sqrt{3}}{2}; \therefore \alpha = \frac{5\pi}{6};$

66. $n(P) = 25\%; n(C) = 15\%; n(P' \cup C') = 65\%;$

$\Rightarrow n(P' \cap C') = 65\%; n(P \cup C) = 35\%;$

$n(P \cap C) = n(P) + n(C) - n(P \cup C) = 25 + 15 - 35 = 5\%;$

$\therefore x \times 5\% = 2000; \Rightarrow x = 40000;$

67. $\frac{\alpha^3}{2}\operatorname{cosec}^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right) + \frac{\beta^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right);$

$\alpha^3 \frac{1}{1-\cos\left(\tan^{-1}\left(\frac{\alpha}{\beta}\right)\right)} + \beta^3 \frac{1}{1+\cos\left(\tan^{-1}\left(\frac{\beta}{\alpha}\right)\right)}$

$\alpha^3 \frac{1}{1-\frac{\beta}{\sqrt{\alpha^2+\beta^2}}} + \beta^3 \frac{1}{1+\frac{\alpha}{\sqrt{\alpha^2+\beta^2}}};$

$\sqrt{\alpha^2+\beta^2}\left\{\alpha^3\left(\frac{\sqrt{\alpha^2+\beta^2}+\beta}{\alpha^2}\right) + \beta^3\left(\frac{\sqrt{\alpha^2+\beta^2}-\alpha}{\beta^2}\right)\right\};$

$\sqrt{\alpha^2+\beta^2}(\alpha+\beta)\sqrt{\alpha^2+\beta^2} = (\alpha+\beta)(\alpha^2+\beta^2);$

68. $A\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}-1\\2\end{bmatrix}$; $A^2\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}$; let A be given by $A = \begin{bmatrix}a & b\\c & d\end{bmatrix}$; The first equation gives $a - b = -1$ *and* $c - d = 2$; For the second equation $A^2\begin{bmatrix}1\\-1\end{bmatrix} = A\left(A\begin{bmatrix}1\\-1\end{bmatrix}\right) = A\left(\begin{bmatrix}-1\\2\end{bmatrix}\right) = \begin{bmatrix}1\\0\end{bmatrix}$; This gives $-a + 2b = 1$ *and* $-c + 2d = 0;$ $(iii) + (iv) \Rightarrow b = 0$ *and* $a = -1;$ $(iv) + (vi) \Rightarrow c = 4$ *and* $d = 2;$ $a + b + c + d = 5,$

69. $y = c_1\cos 2x + c_2\cos^2 x + c_3\sin^2 x + c_4$;

$\Rightarrow c_1\cos 2x + \frac{c_2}{2}(1+\cos 2x) + \frac{c_3}{2}(1-\cos 2x) + c_4;$

$\Rightarrow \left(\frac{c_2}{2} + \frac{c_3}{2} + c_4\right) + \left(c_1 + \frac{c_2}{2} - \frac{c_3}{2}\right)\cos 2x$; $\Rightarrow A + B\cos 2x$;

That means there are two independent parameters. Thus, the orders of differential equation will be 2.

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70. Total no of ways – no of ways start with zero

$\therefore \frac{6!}{3!}-\frac{5!}{2!}=60$

71. Let $X=\begin{bmatrix}a\\b\\c\end{bmatrix}; AX=\lambda X \Rightarrow \begin{bmatrix}8&-6&2\\-6&7&-4\\2&-4&3\end{bmatrix}\begin{bmatrix}a\\b\\c\end{bmatrix}=\begin{bmatrix}\lambda a\\\lambda b\\\lambda c\end{bmatrix}$; $(8-\lambda)a-6b+2c=0$; $-6a+(7-\lambda)b-4c=0$; and $2a-4b+(3-\lambda)c=0$; For non-zero solution $\begin{vmatrix}8-\lambda&-6&2\\-6&7-\lambda&-4\\2&-4&3-\lambda\end{vmatrix}=0$;

$\Rightarrow \lambda=0,3,15$

72. $(1+7)^{83}+(7-1)^{83}=(1+7)^{83}-(1-7)^{83}$

$=2\left[{}^{83}C_1 \,.\, 7+{}^{83}C_3 .7^3+....+{}^{83}C_{83}\,.7^{83}\right]$

$=(2.7\,.\,83)+49I$ where I an integer

Now $14\times 83=1162$

$\therefore \frac{1162}{49}=23\frac{35}{49}$

$\therefore$ Remainder is 35

73. $x^2-\left(3+2^{\sqrt{\log_2^3}}-3^{\sqrt{\log_2^3}}\right)x$

$-2\left(3^{\log_3{}^2}-2^{\log_2{}^3}\right)=0$

$\Rightarrow x^2-3(x)-2(2-3)=0 \qquad \Rightarrow x^2-3x+2=0$

$\Rightarrow \alpha=2,\ \beta=1 \qquad \Rightarrow \alpha^2+\alpha\beta+\beta^2=4+2+1=7$

74. $\frac{4000}{a_1a_{4001}}10 \Rightarrow a_1a_{4001}=400$

$a_1+a_{4001}=50$

$\left(a_1-a_{4001}\right)^2=\left(a_1+a_{4001}\right)^2-4a_1\,a_{4001} \Rightarrow \left|a_1-a_{4001}\right|=30$

75. $\lim_{x\to 0}\frac{x^3}{\sqrt{a}\ (bx-\sin x)}=1; \Rightarrow \lim_{h\to 0}\frac{3x^2}{\sqrt{a}\ (b-\cos x)}=1; \therefore b=1; \lim_{x\to 0}\frac{3x^2}{\sqrt{a}\ \sin^2\frac{x}{2}}$

$=\lim_{x\to 0}\frac{6}{\sqrt{a}}=\frac{6}{\sqrt{a}}; \therefore a=36$;

Thus $a+b=37;$

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